

Appendix C. Deriving Health Impact Functions

This Appendix provides of an overview regarding the health impact functions that BenMAP uses to estimate the impact of a change in air pollution on adverse health effects. It provides a description of the particular types of health impact functions that BenMAP uses.

The functional form of the relationship between the change in pollutant concentration, Δx , and the change in population health response (usually an incidence rate), Δy depends on the functional form of the C-R function from which it is derived, and this depends on the underlying relationship assumed in the epidemiological study chosen to estimate a given effect. For expository simplicity, the following subsections refer simply to a generic adverse health effect, “y” and uses particulate matter (PM) as the pollutant - that is, $\Delta x = \Delta \text{PM}$ - to illustrate how the relationship between Δx and Δy is derived from each of several different C-R functions.

Estimating the relationship between ΔPM and Δy can be thought of as consisting of three steps:

- (1) choosing a functional form of the relationship between PM and y (the C-R function),
- (2) estimating the values of the parameters in the C-R function assumed, and
- (3) deriving the relationship between ΔPM and Δy (the health impact function) from the relationship between PM and y (the C-R function).

Epidemiological studies have used a variety of functional forms for C-R functions. Some studies have assumed that the relationship between adverse health and pollution is best described by a linear form, where the relationship between y and PM is estimated by a linear regression in which y is the dependent variable and PM is one of several independent variables. Log-linear regression and logistic regression are other common forms.

Note that the log-linear form used in the epidemiological literature is often referred to as “Poisson regression” because the underlying dependent variable is a count (e.g., number of deaths), believed to be Poisson distributed. The model parameters may be estimated by regression techniques but are often estimated by maximum likelihood techniques. The form of the model, however, is still log-linear.

C.1 Overview

The relationship between the concentration of a pollutant, x, and the population response, y, is called the concentration-response (C-R) function. For example, the concentration of fine particulate matter (PM_{2.5}) may be in $\mu\text{g}/\text{m}^3$ per day, and the population response may be the number of premature deaths per 100,000 population per day. C-R functions are estimated in epidemiological studies. A functional form is chosen by the researcher, and the parameters of the function are estimated using data

on the pollutant (e.g., daily levels of PM_{2.5}) and the health response (e.g., daily mortality counts). There are several different functional forms, discussed below, that have been used to estimate C-R functions. The one most commonly used is the log-linear form, in which the natural logarithm of the health response is a linear function of the pollutant concentration.

For the purposes of estimating benefits, we are not interested in the C-R function itself, however, but the relationship between the change in concentration of the pollutant, Δx , and the corresponding change in the population health response, Δy . We want to know, for example, if the concentration of PM_{2.5} is reduced by 10 $\mu\text{g}/\text{m}^3$, how many premature deaths will be avoided? The relationship between Δx and Δy can be derived from the C-R function, as described below, and we refer to this relationship as a health impact function.

Many epidemiological studies, however, do not report the C-R function, but instead report some measure of the change in the population health response associated with a specific change in the pollutant concentration. The most common measure reported is the relative risk associated with a given change in the pollutant concentration. A general relationship between Δx and Δy can, however, be derived from the relative risk. The relative risk and similar measures reported in epidemiological studies are discussed in the sections below. The derivation of the relationship of interest for BenMAP - the relationship between Δx and Δy - is discussed in the subsequent sections.

C.2 Review Relative Risk and Odds Ratio

The terms relative risk and odds ratio are related but distinct. Table C-1 provides the basis for demonstrating their relationship.

Table C-1. Relative Risk and Odds Ratio Notation

Exposure	Fraction of Population		Adverse Effect Measure	
	Affected	Not Affected	Relative Risk	Odds
Baseline Pollutant Exposure	y_0	$1-y_0$	y_0/y_c	$y_0/(1-y_0)$
Control Pollutant Exposure	y_c	$1-y_c$		$y_c/(1-y_c)$

The “risk” that people with baseline pollutant exposure will be adversely affected (e.g., develop chronic bronchitis) is equal to y_0 , while people with control pollutant exposure face a risk, y_c , of being adversely affected. The relative risk (RR) is simply:

$$RR = \frac{y_0}{y_c}$$

The odds that an individual facing high exposure will be adversely affected is:

$$Odds = \frac{y_0}{1 - y_0}$$

The odds ratio is then:

$$Odds\ Ratio = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)}$$

This can be rearranged as follows:

$$Odds\ Ratio = \frac{y_0}{y_c} \times \left(\frac{1-y_c}{1-y_0}\right) = RR \times \left(\frac{1-y_c}{1-y_0}\right)$$

As the risk associated with the specified change in pollutant exposure gets small (i.e., both y_0 and y_c approach zero), the ratio of $(1-y_c)$ to $(1-y_0)$ approaches one, and the odds ratio approaches the relative risk. This relationship can be used to calculate the pollutant coefficient in the C-R function from which the reported odds ratio or relative risk is derived, as described below.

C.3 Linear Model

A linear relationship between the rate of adverse health effects (incidence rate) and various explanatory variables is of the form:

$$y = \alpha + \beta \times PM$$

where α incorporates all the other independent variables in the regression (evaluated at their mean values, for example) times their respective coefficients. The relationship between the change in the rate of the adverse health effect from the baseline rate (y_0) to the rate after control (y_c) associated with a change from PM_0 to PM_c is then:

$$\Delta y = y_0 - y_c = \beta * (PM_0 - PM_c) = \beta * \Delta PM$$

For example, Ostro et al. (1991, Table 5) reported a $PM_{2.5}$ coefficient of 0.0006 (with a standard error of 0.0003) for a linear relationship between asthma and $PM_{2.5}$ exposure.

The lower and upper bound estimates for the $PM_{2.5}$ coefficient are calculated as follows:

$$\beta_{lowerbound} = \beta - (1.96 \times \sigma_\beta) = 0.0006 - (1.96 \times 0.0003) = 1.2 \times 10^{-5}$$

$$\beta_{upperbound} = \beta + (1.96 \times \sigma_\beta) = 0.0006 + (1.96 \times 0.0003) = 0.00119$$

It is then straightforward to calculate lower and upper bound estimates of the change in asthma.

C.4 Log-linear Model

The log-linear relationship defines the incidence rate (y) as:

$$y = B \times e^{\beta \cdot PM}$$

Or, equivalently,

$$\ln(y) = \alpha + \beta \cdot PM,$$

where the parameter B is the incidence rate of y when the concentration of PM is zero, the parameter β is the coefficient of PM, $\ln(y)$ is the natural logarithm of y, and $\alpha = \ln(B)$. Other covariates besides pollution clearly affect mortality. The parameter B might be thought of as containing these other covariates, for example, evaluated at their means. That is,

$$B = B_0 \times e^{\beta_1 x_1 + \dots + \beta_n x_n}$$

where B_0 is the incidence of y when all covariates in the model are zero, and $x_1, \dots, x_n$ are the other covariates evaluated at their mean values. The parameter B drops out of the model, however, when changes in y are calculated, and is therefore not important.

The relationship between ΔPM and Δy is:

$$\Delta y = y_0 - y_c = B(e^{\beta PM_0} - e^{\beta PM_c})$$

This may be rewritten as:

$$\Delta y = B \times e^{\beta \cdot PM_0} (1 - e^{-\beta(PM_0 - PM_c)}) = y_0 \left(1 - \frac{1}{\exp(\beta \times \Delta PM)} \right)$$

where y_0 is the baseline incidence rate of the health effect (i.e., the incidence rate before the change in PM).

The change in the incidence of adverse health effects can then be calculated by multiplying the change in the incidence rate, Δy , by the relevant population (e.g., if the rate is number per 100,000 population, then the relevant population is the number of 100,000s in the population).

When the PM coefficient (β) and its standard error (σ_β) are published (e.g., Ostro et al., 1989), then the coefficient estimates associated with the lower and upper bound may be calculated easily as follows:

$$\beta_{lowerbound} = \beta - (1.96 \times \sigma_\beta)$$

$$\beta_{upperbound} = \beta + (1.96 \times \sigma_\beta),$$

Where the adjustment on the mean of ± 1.96 times the standard error produces the 2.5th and 97.5th percentiles of the normal distribution, which are used to approximate a 95%

confidence interval. These values can be changed to capture different lower and upper bounds.

Epidemiological studies often report a relative risk for a given ΔPM , rather than the coefficient, β (e.g., Schwartz et al., 1995, Table 4). Recall that the relative risk (RR) is simply the ratio of two risks:

$$RR = \frac{y_0}{y_c} = e^{\beta \cdot \Delta PM}$$

Taking the natural log of both sides, the coefficient in the C-R function underlying the relative risk can be derived as:

$$\beta = \frac{\ln(RR)}{\Delta PM}$$

The coefficients associated with the lower and upper bounds (e.g., the 2.5th and 97.5th percentiles) can be calculated by using a published confidence interval for relative risk, and then calculating the associated coefficients.

Because of rounding of the published RR and its confidence interval, the standard error for the coefficient implied by the lower bound of the RR will not exactly equal that implied by the upper bound, so an average of the two estimates is used. The underlying standard error for the coefficient (σ_β) can be approximated by:

$$\begin{aligned}\sigma_{\beta, 2.5 \text{ percentile}} &= \frac{\beta - \beta_{2.5 \text{ percentile}}}{1.96} \\ \sigma_{\beta, 97.5 \text{ percentile}} &= \frac{\beta_{97.5 \text{ percentile}} - \beta}{1.96} \\ \sigma_\beta &\cong \frac{\sigma_{\beta, 2.5 \text{ percentile}} + \sigma_{\beta, 97.5 \text{ percentile}}}{2}\end{aligned}$$

C.5 Logistic Model

In some epidemiological studies, a logistic model is used to estimate the probability of an occurrence of an adverse health effect. Given a vector of explanatory variables, X , the logistic model assumes the probability of an occurrence is:

$$y = \text{prob}(\text{occurrence} \mid X \times \beta) = \left(\frac{e^{X \cdot \beta}}{1 + e^{X \cdot \beta}} \right),$$

where β is a vector of coefficients. Greene (1997, p. 874) presents models with discrete dependent variables, such as the logit model. See also Judge et al. (1985, p. 763). This may be rewritten as:

$$y = \frac{e^{X \cdot \beta}}{1 + e^{X \cdot \beta}} \times \frac{e^{-X \cdot \beta}}{e^{-X \cdot \beta}} = \frac{1}{1 + e^{-X \cdot \beta}}$$

The odds of an occurrence is:

$$\begin{aligned} odds &= \frac{y}{1-y} = \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{1 - \frac{1}{1 + e^{-X \cdot \beta}}} \\ \Rightarrow odds &= \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{1 - \frac{1}{1 + e^{-X \cdot \beta}}} = \frac{\left(\frac{1}{1 + e^{-X \cdot \beta}} \right)}{\left(\frac{e^{-X \cdot \beta}}{1 + e^{-X \cdot \beta}} \right)} = \frac{1}{e^{-X \cdot \beta}} = e^{X \cdot \beta} \\ \Rightarrow \ln(odds) &= X \times \beta \end{aligned}$$

The odds ratio for the control scenario ($odds_c$) versus the baseline ($odds_0$) is then:

$$odds\ ratio = \frac{odds_c}{odds_0} = \frac{\left(\frac{y_c}{1-y_c} \right)}{\left(\frac{y_0}{1-y_0} \right)} = \frac{\left(\frac{1}{e^{-X_c \cdot \beta}} \right)}{\left(\frac{1}{e^{-X_0 \cdot \beta}} \right)} = \frac{e^{X_c \cdot \beta}}{e^{X_0 \cdot \beta}}$$

The *change* in the probability of an occurrence from the baseline to the control (Δy), assuming that all of the other covariates remain constant, may be derived from this odds ratio:

$$\begin{aligned} odds\ ratio &= \frac{\frac{y_0}{1-y_0}}{\frac{y_c}{1-y_c}} = \frac{e^{X_0 \cdot \beta}}{e^{X_c \cdot \beta}} = e^{\beta \cdot \Delta X} = e^{\beta \cdot \Delta PM} \\ \frac{y_c}{1-y_c} &= \frac{y_0}{1-y_0} \times e^{-\beta \cdot \Delta PM} \end{aligned}$$

$$y_c = (1 - y_0) \times \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c + y_c \times \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM} = \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c \left(1 + \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM} \right) = \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}$$

$$y_c = \frac{\frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}}{1 + \frac{y_0}{1 - y_0} \times e^{-\beta \cdot \Delta PM}} = \frac{y_0 \times e^{-\beta \cdot \Delta PM}}{1 - y_0 + y_0 \times e^{-\beta \cdot \Delta PM}} = \frac{y_0}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0}$$

$$\Delta y = y_0 - y_c = y_0 - \frac{y_0}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0}$$

$$\Delta Incidence = \Delta y \times pop = y_0 \times \left(1 - \frac{1}{(1 - y_0) \times e^{\beta \cdot \Delta PM} + y_0} \right) \times pop$$

When the coefficient (β) and its standard error (σ_β) are published (e.g., Pope et al., 1991, Table 5), then the coefficient estimates associated with the lower and upper bound may be calculated easily as follows:

$$\beta_{lowerbound} = \beta - (1.96 \times \sigma_\beta)$$

$$\beta_{upperbound} = \beta + (1.96 \times \sigma_\beta),$$

where the adjustments to the mean of plus or minus 1.96 times the standard error represent the 2.5th and 97.5th percentiles of the normal distribution, and are used to approximate a 95% confidence interval. These values can be changed to capture different lower and upper bounds.

Often the logistic regression coefficients are not published, and only the odds ratio corresponding to a specified change in PM is presented (e.g., Schwartz et al., 1994). It is easy to calculate the underlying coefficient as follows:

$$\ln(odds\ ratio) = \beta \times \Delta PM$$

$$\Rightarrow \beta = \frac{\ln(odds\ ratio)}{\Delta PM}$$

The coefficients associated with the lower and upper bound estimates of the odds ratios are calculated analogously. The underlying standard error for the coefficient (σ_β) can be approximated by:

$$\sigma_{\beta, 2.5\ percent} = \frac{\beta - \beta_{2.5\ percent}}{1.96}$$

$$\sigma_{\beta, 97.5\ percent} = \frac{\beta_{97.5\ percent} - \beta}{1.96}$$

$$\sigma_\beta \cong \frac{\sigma_{\beta, 2.5\ percent} + \sigma_{\beta, 97.5\ percent}}{2}$$

Sometimes, however, the relative risk is presented. The relative risk does not equal the odds ratio, and a different procedure should be used to estimate the underlying coefficient. Note that ESEERCO (1994, p. V-21) calculated (incorrectly) the underlying regression coefficient for Abbey et al. (1993, Table 5) by taking the logarithm of the relative risk and dividing by the change in TSP.

The relative risk (RR) is simply:

$$RR = \frac{y_0}{y_c},$$

where y_0 is the risk (i.e., probability of an occurrence) at the baseline PM exposure and y_c is the risk at the control PM exposure. When the baseline incidence rate (y_0) is given, then it is easy to solve for the control incidence rate (y_c):

$$y_c = \frac{y_0}{RR},$$

The odds ratio, may then be calculated:

$$odds\ ratio = \frac{\frac{y_0}{1 - y_0}}{\frac{y_c}{1 - y_c}}$$

Given the odds ratio, the underlying coefficient (β) may be calculated as before:

$$\beta = \frac{\ln(odds\ ratio)}{\Delta PM},$$

The odds ratio and the coefficient calculated from it are dependent on the baseline and control incidence rates. Unfortunately, it is not always clear what the baseline and control incidence rates should be. Abbey et al. (1995b, Table 2) reported that there are 117 new cases of chronic bronchitis out of a sample of 1,631, or a 7.17 percent rate. In addition, they reported the relative risk (RR = 1.81) for a new case of chronic bronchitis associated with an annual mean concentration “increment” of 45 $\mu\text{g}/\text{m}^3$ of $\text{PM}_{2.5}$ exposure.

Assuming that the baseline rate for chronic bronchitis (y_0) should be 7.17 percent, the question becomes whether the “increment” of 45 $\mu\text{g}/\text{m}^3$ should be added to or subtracted from the existing $\text{PM}_{2.5}$ concentration. If added, the control incidence rate (y_c) would be greater than the baseline rate (y_0), while subtraction would give a control rate less than the incidence rate. In effect, one might reasonably derive two estimates of the odds ratio:

$$odds\ ratio_1 = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)} = \frac{\left(\frac{1.81 \times 0.0717}{1-(1.81 \times 0.0717)}\right)}{\left(\frac{0.0717}{1-0.0717}\right)} = 1.931$$

$$odds\ ratio_2 = \frac{\left(\frac{y_0}{1-y_0}\right)}{\left(\frac{y_c}{1-y_c}\right)} = \frac{\left(\frac{0.0717}{1-0.0717}\right)}{\left(\frac{0.0717}{1-\frac{0.0717}{1.81}}\right)} = 1.873$$

$$\Rightarrow \beta_1 = \frac{\ln(1.931)}{45} = 0.01462$$

$$\Rightarrow \beta_2 = \frac{\ln(1.873)}{45} = 0.01394$$

An alternative is to simply assume that the relative risk (1.81) is reasonably close to the odds ratio and calculate the underlying coefficient. It is easy to show that the relative risk equals:

$$RR = \frac{y_0}{y_c} = (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0$$

Assuming that:

$$e^{-\Delta PM \cdot \beta} \cong (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0$$

$$\Rightarrow RR \cong e^{-\Delta PM \cdot \beta}$$

It is then possible to calculate the underlying coefficient:

$$\frac{\ln(RR)}{-\Delta PM} \cong \beta$$

$$\Rightarrow \beta_3 = \frac{\ln(1.81)}{45} = 0.01319$$

Since this coefficient estimate is based on the assumption that

$$e^{-\Delta PM \cdot \beta} \cong (1 - y_0) \times e^{-\Delta PM \cdot \beta} + y_0 ,$$

it should be used in a C-R function that maintains this assumption. In effect, it should be applied to a log-linear C-R function:

$$\Delta y = \left[y_0 \times \left(e^{\beta \cdot \Delta PM} - 1 \right) \right]$$

Using the formula for the change in the incidence rate and assuming a 10 µg/m³ decline in PM_{2.5}, it is shown that this results in changes within the bounds suggested by the two estimates based on using the estimated odds ratios:

$$\Delta y_1 = \frac{.0717}{(1 - 0.0717) \times e^{10 \times 0.01462} + 0.0717} - 0.0717 = -0.00914$$

$$\Delta y_2 = \frac{.0717}{(1 - 0.0717) \times e^{10 \times 0.01394} + 0.0717} - 0.0717 = -0.00874$$

$$\Delta y_3 = 0.0717 \times \left(e^{-10 \times 0.01319} - 1 \right) = -0.00886$$

In this instance, it seems that simply using the relative risk to estimate the underlying coefficient results in a good approximation of the change in incidence. Since it is unclear which of the two other coefficients (β_1 or β_2) should be used - as the published work was not explicit - the coefficient based on the relative risk and the log-linear functional form is a reasonable approach.

C.6 Cox Proportional Hazards Model

Use of a Cox proportional hazards model in an epidemiological study results in a C-R function that is log-linear in form. It is often used to model survival times, and as a result, this discussion focuses on mortality impacts.

The Cox proportional hazards model is based on a hazard function, defined as the probability that an individual dies at time t , conditional on having survived up to time t (Collet, 1994, p. 10). More formally, the hazard function equals the probability density function for the risk of dying divided by one minus the cumulative probability density function:

$$h(X, t) = \frac{f(X, t)}{1 - F(X, t)}$$

The proportional hazards model takes the form:

$$h(X, t) = h_0(t) e^{X \cdot \beta},$$

where X is a vector of explanatory variables, β is a vector of coefficients, and $h_0(t)$ is the so-called “baseline hazard” rate. This terminology differs from that used in most of this discussion: this “baseline hazard” is the risk when all of the covariates (X) are set to zero; this is *not* the risk in the baseline scenario.

The Cox proportional hazards model is sometimes termed a “semi-parametric” model, because the baseline hazard rate is calculated using a non-parametric method, while the impact of explanatory variables is parameterized. Collet (1994) details the estimation of Cox proportional hazards models; in particular, see Collet’s discussion (pp. 95-97) of nonparametric estimation of the baseline hazard.

Taking the ratio of the hazard functions for the baseline and control scenarios gives the relative risk:

$$RR = \frac{h(X_0, t)}{h(X_c, t)} = \frac{h_0(t) e^{X_0 \cdot \beta}}{h_0(t) e^{X_c \cdot \beta}} = e^{\Delta PM \cdot \beta},$$

where it is assumed that the only difference between the baseline and control is the level of PM pollution.

The relative risk is often presented rather than the coefficient β , so it is necessary to estimate β in order to develop the functional relationship between ΔPM and Δy , as described previously for log-linear C-R functions.